

HE2004 Introductory Econometrics: Linear Regression with One Regressor 5

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Tests and Confidence Intervals with a Single Regressor

KEY CONCEPT: t-statistic for the OLS Estimator

- The t -statistic for $\hat{\beta}_1$ can be written as:

$$t = \frac{\hat{\beta}_1 - \beta_{1,0}}{SE(\hat{\beta}_1)} = \frac{\text{estimator} - \text{hypothesized value}}{\text{standard error of the estimator}}.$$

where $\beta_{1,0}$ is the value tested by the null hypothesis.

- Especially, if the null hypothesis is $H_0 : \beta_1 = 0$, then**

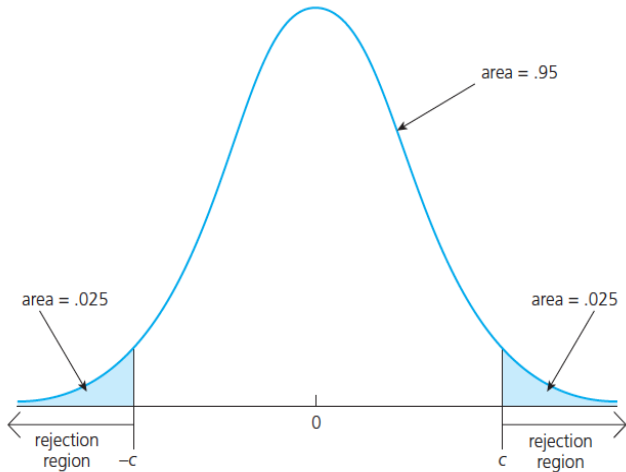
$$t = \frac{\hat{\beta}_1}{SE(\hat{\beta}_1)}.$$

- Because the t -statistic follows $N(0, 1)$ in large samples, the null hypothesis is rejected at the 5% significance level if $|t^{act}| > 1.96$. Here, t^{act} refers to the t -statistic computed using the **actual data**: That is, computed using $\hat{\beta}_1$ and $SE(\hat{\beta}_1)$ with the actual data.

Similarly, one may use this procedure to test the value of β_0 .

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5% significance level t-test against the two-sided alternative: The critical value c is equal to 1.96.



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A Big Picture View of What We Are Doing

We want to learn about the slope of the population regression line. **We have data from a sample (not the population)**, so there is **sampling uncertainty** (if the sample is changed, our estimate will also change). Therefore, we want **an estimate of the slope**, as well as **a measure of the associated uncertainty**.

There are five steps towards our goal:

1. State the model and (population) **object of interest** (e.g., β_1);
2. Provide an **estimator** of this object (e.g., the OLS estimator $\hat{\beta}_1$);
3. Study the **distribution of $\hat{\beta}_1$** . In large samples (and if the OLS assumptions are satisfied), this distribution is approximately **normal with mean equal to the true value β_1** ;
4. Compute the **standard error $SE(\hat{\beta}_1)$** ;
5. Construct the **t-statistic** for hypothesis testing.

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Application to the Test Score Data

- Because standard errors are very important for statistical analysis (measure of **sampling uncertainty**), they are always reported with OLS coefficients.
- A common format to report the standard errors is to place them in **parentheses below** the respective coefficients of the regression line:

$$\widehat{TestScore} = 689.9 - 2.28 \times STR.$$

(10.4) (0.52)

- This means that $SE(\hat{\beta}_0) = 10.4$ and $SE(\hat{\beta}_1) = 0.52$. In this example, we can test $H_0 : \beta_1 = 0$ (i.e., **no effect of class size** on student performance) by using

$$t^{act} = (-2.28 - 0)/0.52 = -4.38.$$

Because $|t^{act}| > 1.96$, H_0 is rejected in favor of the two-sided alternative (i.e., $H_1 : \beta_1 \neq 0$) at the 5% significance level.

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```
. reg testscr str, robust
```

Linear regression

Number of obs = 420
F(1, 418) = 19.26
Prob > F = 0.0000
R-squared = 0.0512
Root MSE = 18.581

testscr	Robust					
	Coef.	Std. Err.	t	P> t	[95% Conf. Interval]	
str	-2.279808	.5194892	-4.39	0.000	-3.300945	-1.258671
_cons	698.933	10.36436	67.44	0.000	678.5602	719.3057

*Note: $t = -4.39$ in STATA due to the difference by rounding.

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KEY CONCEPT: Confidence Intervals for a Regression Coefficient

- A 95% (two-sided) confidence interval for β_1 is an interval that contains the true value of β_1 with a 95% probability.
- Equivalently, it is the set of values that cannot be rejected by a (two-sided) hypothesis test with a 5% significance level.
- Let CI_{β_1} denote the 95% confidence interval of β_1 . Then

$$\begin{aligned} CI_{\beta_1} &= \left\{ \beta_1 : \left| \frac{\hat{\beta}_1 - \beta_1}{SE(\hat{\beta}_1)} \right| \leq 1.96 \right\} \\ &= \left[\hat{\beta}_1 - 1.96 \times SE(\hat{\beta}_1), \hat{\beta}_1 + 1.96 \times SE(\hat{\beta}_1) \right]. \end{aligned}$$

A confidence interval of β_0 can be obtained in a similar way.

Tests and Confidence Intervals with a Single Regressor

Application to the Test Score Data (Cont'd)

$$\widehat{TestScore} = 689.9 - 2.28 \times STR.$$

(10.4) (0.52)

- The 95% confidence interval for β_1 is

$$[-2.28 - 1.96 \times 0.52, -2.28 + 1.96 \times 0.52],$$

that is, $-3.30 \leq \beta_1 \leq -1.26$. Thus, there is **approximately 95% probability** that the **true β_1 is in this interval** (“approximately” because we used the approximation of normal distribution).

- Note that the value **$\beta_1 = 0$ is not contained in this confidence interval**, which is **equivalent to** the fact that $H_0 : \beta_1 = 0$ can be rejected at the 5% significance level (remember $t^{act} = -4.38$ in this case); This result suggests that the **class size does have some effect** on students' performance, based on the current data.

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KEY CONCEPT: p-value

The **p-value** is often reported with the **t-statistic and confidence interval**: It refers to the probability of observing a t-statistic **at least as extreme as** what we obtained in data (-4.38 in the example), **given that H_0 is true**.

- For the (two-sided) t-test using normal distribution, we have:

$$\text{p-value} = P_{H_0} (|N(0, 1)| > |t^{act}|)$$

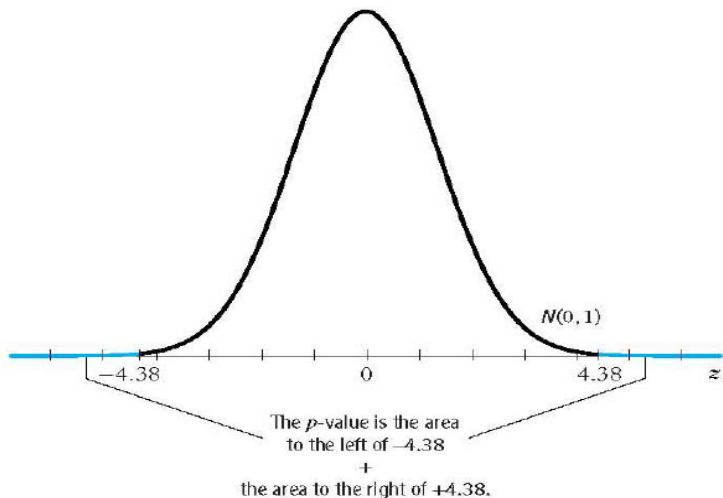
where $N(0, 1)$ follows a standard normal distribution, while t^{act} is the actual value of the t-statistic computed using your data.

- Therefore, the p-value computes the probability that $N(0, 1)$, a **standard normal random variable**, gets a value that is **at least as extreme as** your t-statistic when the null hypothesis is true.
- A **small p-value provides evidence against the null hypothesis H_0** .

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- In particular, if the **p-value is less than 5%**, we can reject the null hypothesis H_0 at the 5% significance level; If the **p-value is less than 1%**, we can reject H_0 at the 1% significance level.
- Large $|t^{act}| \rightarrow$ **Small p-value**; Small $|t^{act}| \rightarrow$ **Large p-value**.
- The p-value for $H_0 : \beta = 0$ in the California data is around 0.000001.
- That is, if H_0 is true, the probability of getting a value of t -statistic as extreme as what we obtained ($t^{act} = -4.38$) is just around 0.000001. $\rightarrow$ Therefore, we conclude that H_0 **should be false**.

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The p -value based on the large- n standard normal approximation to the t -statistic is 0.00001 (10^{-5})

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Empirical Application: CEO Salary and Return on Equity (Cont'd)

- $\widehat{salary} = 963.191 + 18.501roe$.
- $\hat{\beta}_1 = 18.501, SE(\hat{\beta}_1) = 6.829$.
- For the slope β_1 ($\Delta salary / \Delta roe$), $t^{act} = 2.71$, p-value = 0.007;
 $H_0 : \beta_1 = 0$ rejected at 5% and 1% significance levels.
- The 95% confidence interval for β_1 is [5.04, 31.97].
- Firm performance (roe) does have significant impact on ceo's salary.

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```
. regress salary roe, robust
```

Linear regression

Number of obs = 209

F(1, 207) = 7.34

Prob > F = 0.0073

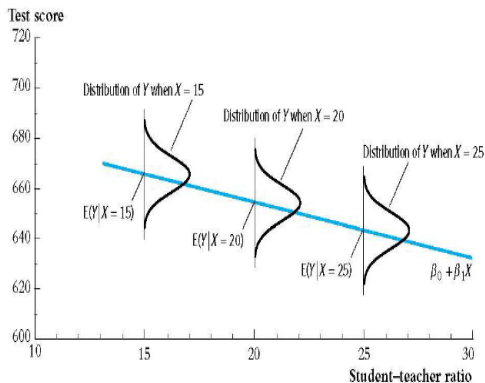
R-squared = 0.0132

Root MSE = 1366.6

salary	Robust					
	Coef.	Std. Err.	t	P> t	[95% Conf. Interval]	
roe	18.50119	6.829448	2.71	0.007	5.036994	31.96538
_cons	963.1913	121.1062	7.95	0.000	724.4316	1201.951

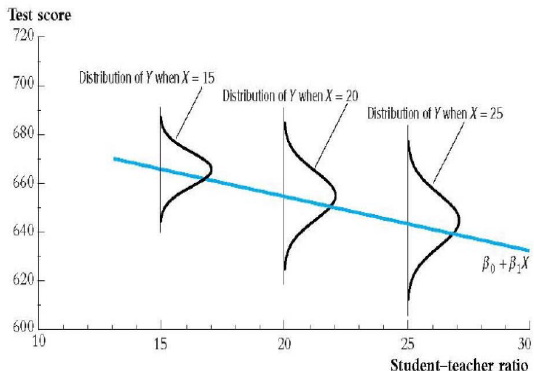
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Homoskedasticity: The distribution of Y (and the distribution of the errors u : Remember that the only difference between Y and u is the population regression line) is shown for different values of X . All these distributions have the same spread; That is, the variance of these distributions is the same for different values of X (15, 20, 25,...). Thus, the errors in this figure are homoskedastic (the variance of u does not change with the value of X).



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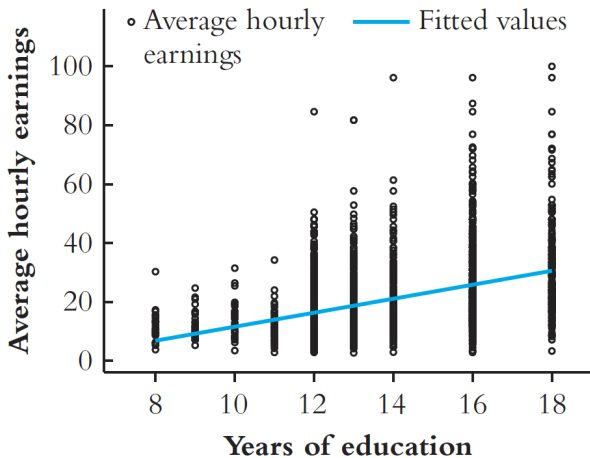
Heteroskedasticity: By contrast, the following figure illustrates a case in which the **distribution of Y** (and the **distribution of u**) spreads out as the value of X increases. For **small values of X** , the **distribution is tight**, but for **large values of X** , the **distribution has a larger spread**. In general, the errors are **heteroskedastic** when the **variance of u** changes with the **value of X** .



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So, how is Heteroskedasticity related to real data?

A real-data example: Earning vs. years of education; 2731 full-time US workers in 2015, ages 29 and 30, with 8 - 18 years of education. The data come from 2016 Current Population Survey by US Census Bureau.



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- The figure has **two striking features**. The first is that the mean of the distribution of earnings increases with the number of years of education. This increase is summarized by the **OLS regression line** (we keep in mind that these estimates can be biased if the OLS assumptions are not satisfied):

$$\widehat{\text{Earnings}} = -12.12 + 2.37 \text{ Years of Education}, \quad (1)$$

(1.36) (0.10)

$$R^2 = 0.185.$$

The corresponding **95% confidence interval** for β_1 is $[2.37 - 2.96 \times 0.10, 2.37 + 1.96 \times 0.10] = [2.17, 2.57]$.

- The second striking feature is that **the spread of the distribution of earnings increases with the year of educations**. While some workers with many years of education have low-paying jobs, very few workers with low levels of education have high-paying jobs.

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Reason for heteroskedasticity in earning?

- On average, workers with more education have higher earning than workers with less education.
- If the **best-paying jobs mainly go to the university graduated** (e.g., bankers, consultants, IT), it may also be that the **spread/dispersion of the distribution of earnings** is greater for workers with more education.
- According to the figure, not all **university graduates** (with around 16 years of education) will be earning \$60 per hour by the time they are 29, but some will, and workers with only **10 years of education have no shot at those jobs**.
- As this example illustrates, heteroskedasticity may arise in many economic applications. In such cases, we may use **heteroskedasticity-robust standard errors** by using the **“robust” option in STATA**; e.g., **reg Y X, robust**.